3.1 General properties of waves — 5054 Paper 2

Newest to oldest (2025 → 2015). Only 3.1 subquestions. Official mark-scheme pages under each clip.

2025

Oct/Nov 5054/22

Question 5 · included (b)(i)

only $v=f\lambda$

(i) Calculate the energy transferred from the gravitational potential energy store of the container.

energy transferred = ... J [2]

Mark scheme — 5054/22 Q5

5(b)(i) (■ =) c f or 8 3.0 10 1.2 10 N ■ ■ C1 2.5 ■ 10X (m) C1 0.025 (m) A1

Question	Answer	Marks
5(a)	| | | UV | | | M | | increasing wavelength →	B1
5(b)(i)	(λ =) $\frac{c}{f}$ **or** $\frac{3.0\times10^{8}}{1.2\times10^{N}}$	C1
	2.5×10^{X} (m)	C1
	0.025 (m)	A1
5(b)(ii)	encoded microwave / microwave signal / microwave carrying data / microwave transmitting data (to satellite from Earth)	B1
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	B1
5(c)	to prevent microwaves escaping / leaving the oven / spreading out of the device	B1
	microwaves / they can cause burns (in living tissue / people) / damage to tissue through heating / blisters	B1

Question	Answer	Marks
6(a)	a d.c. / it has one direction **or** does not (repeatedly) reverse direction **or** polarity / direction does not change	B1
6(b)(i)	(Q =) It **or** 41×120	C1
	4.9×10^{3} (C)	A1

May/June 5054/21

Question 5 · included (a), (b), (c)

(a) Describe the transfers between energy stores that occur as the load is lifted.

....

....

....

.... [3]

(b) The efficiency of the motor, pulley and load system is less than 100%.

(c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

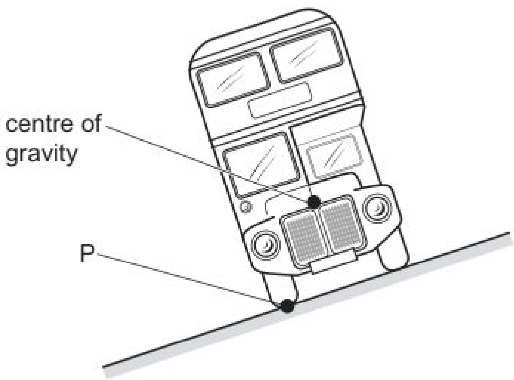

Fig. 2.2

Mark scheme — 5054/21 Q5

5(a)(i) C and R marked correctly B1 5(a)(ii) compressions are where pressure is high / particles close together / density is high B1 5(b)(i) 0.75 (Hz) B1 5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 15 Question Answer Marks

Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	*speed:* no change	B1
	wavelength: decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other **and** on correct side of normal	M1
	refracted towards the surface on correct side of surface **and** join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle **or** spreading out after passing through a (small) gap / hole	B1

May/June 5054/22

Question 4 · included (a), (b), (c)

(a) Define 'power'.

....

.... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ... W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

Mark scheme — 5054/22 Q4

4(a) motion of particles: backwards and forwards / to and fro / closer and further apart B1 (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions B1 4(b)(i) number of oscillations per second or number of wave(length)s (passing a point) per second B1 4(c)(i) (at least) three correctly curved wavefronts centered on gap B1 same wavelength as on left and some curving in correct direction B1

Question	Answer	Marks
4(a)	*motion of particles:* backwards and forwards / to and fro / closer and further apart	**B1**
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork **or** mention of compressions and rarefactions	**B1**
4(b)(i)	number of oscillations per second **or** number of wave(length)s (passing a point) per second	**B1**
4(b)(ii)	2.5 (waves) seen **or** (1 wave takes) 0.02 (s) **or** (f =) 1 / T or (number of) waves / time	**C1**
	50 (Hz)	**A1**
4(b)(iii)	quieter / less volume	**B1**
4(b)(iv)	half frequency shown **and** at least one complete oscillation seen	**B1**
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	**B1**
	same wavelength as on left **and** some curving in correct direction	**B1**
4(c)(ii)	diffraction	**B1**

2024

Oct/Nov 5054/21

Question 5 · included (a), (b), (c)

(a) Explain, in terms of electrons and the atoms of the metal, how thermal energy is transferred through the walls of the metal cylinder.

....

....

....

....

....

.... [3]

(b) The piston is fixed in position so the pressure of the gas increases as the temperature of the gas increases.

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(c) The gas in the cylinder reaches a temperature of 100 °C.

The piston is released and moves to the right. The temperature remains at 100 °C.

The pressure of the gas now decreases.

(i) Explain, in terms of the particles of the gas, why the pressure of the gas now decreases.

....

....

.... [1]

(ii) Eventually, the piston stops moving.

Explain why.

....

....

....

.... [2]

[Total: 10]

Mark scheme — 5054/21 Q5

5(a) ultrasound / sound / it is transmitted by vibrating matter / particles B1 no matter / particles in a vacuum / no medium B1 5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1 5(c)(i) (■ =) v / f or 1500 / 8.4 ■ 106 C1 1.8 ■ 10N C1 1.8 ■ 10–4 (m) A1

Question	Answer	Marks
4(c)(ii)	frequency increases **and** wavelength decreases	**B1**
4(d)	infrared (radiation)	**B1**

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by vibrating matter / particles	**B1**
	no matter / particles in a vacuum / no medium	**B1**
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	**B1**
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	**B1**
5(c)(i)	(λ =) v / f **or** 1500 / 8.4 × 10^6	**C1**
	1.8 × 10^N	**C1**
	1.8 × 10^{-4} (m)	**A1**
5(c)(ii)	frequency: does not change	**B1**
	wavelength: increases	**B1**

May/June 5054/21

Question 4 · included (a)(i), (a)(ii), (b), (c)

not (a)(iii) ultrasound use

(a) **(i)** On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

(b) The speeds of the ball at A, B, C and D are v_A, v_B, v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest ——→—— fastest

[1]

(c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

....

....

....

.... [3]

Mark scheme — 5054/21 Q4

4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) (V =) f■■ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) (V =) f■■ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1 4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

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Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	B1

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	B1
4(a)(ii)	(V =) $f\lambda$ in any form	C1
	330 m / s	A1
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	B1
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	B1
	transverse wave - oscillation at right angles to (direction of travel of) wave	B1
	diagram of each type	B1
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	B1
4(c)	long wavelengths bend around hill **and** short wavelengths do not bend	B1
	(due to) diffraction	B1

Question	Answer	Marks
5(a)	correct symbol; 	B1
5(b)(i)	correct curvature	B1
5(b)(ii)	(at higher current / voltage) temperature increases	B1
	(at higher current / voltage) resistance increases	B1

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2023

Oct/Nov 5054/21

Question 4 · included (a), (b)(i)

not n or ray path

(a) Define what is meant by 'acceleration'.

....

....

.... [2]

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

....

.... [1]

Mark scheme — 5054/21 Q4

4(a) (■ =) v / f or 3.0 ■ 108 / 4.7 ■ 1014 C1 6.4 ■ 10N (m) or 3 / 4.7 ■ 106 C1 6.4 ■ 10–7 (m) A1 4(b)(i) speed: decreases frequency: stays the same wavelength: decreases B2

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Question	Answer	Marks
4(a)	(λ =) *v* / *f* **or** 3.0 × 10^8 / 4.7 × 10^{14}	C1
	6.4 × 10^N (m) or 3 / 4.7 × 10^6	C1
	6.4 × 10^{-7} (m)	A1
4(b)(i)	speed: decreases	B2
	frequency: stays the same	
	wavelength: decreases	
4(b)(ii)	*i* = 45° **and** *r* = 30°	C1
	(*n* =) sin(*i*) / sin(*r*) **or** sin(45°) / sin(30°)	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	B1
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

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Question 5 · included (b)

only trans vs long

(b) The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17 °C and the heater is switched on.

Mark scheme — 5054/21 Q5

5(b) in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction B1 in a longitudinal wave, the vibration direction is parallel to the propagation direction B1 Question Answer Marks

Question	Answer	Marks
4(a)	(λ =) v / f **or** $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	i = 45° **and** r = 30°	C1
	(n =) sin(i) / sin(r) **or** sin(45°) / sin(30°)	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	B1
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

Question	Answer	Marks
5(b)	in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the vibration direction is parallel to the propagation direction	B1

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	(R =) V / I **or** 1.5 / 0.20 **or** 7.5 (Ω)	C1
	(R_T =) 7.5 – 1.5 –3.5	C1
	2.5 (Ω)	A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases **or** smaller fraction of total resistance	B1
	p.d. across R and S increases **or** (p.d. is) smaller fraction of e.m.f.	B1

Question	Answer	Marks
7(a)(i)	P: (carbon) brush **and** Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X **and** no indications elsewhere	B1

Oct/Nov 5054/22

Question 10 · included (b)

only v=fλ

(b) The weight produces a moment about the pivot through H.

Mark scheme — 5054/22 Q10

10(b) (■ =) v / f = 3.0 ■ 108 / 8.8 ■ 1019 C1 3.4 ■ 10–12 (m) A1

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Question	Answer	Marks
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	B1
	outward force (due to high temperature / nuclear reaction) **or** force due to high temperature / nuclear reaction **or** radiation pressure	B1
	balances (gravitational) force (inwards)	B1
9(c)	(red supergiant explodes as) supernova (explosion)	B1

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ **and** no other path indicated as γ	B1
10(a)(ii)	beta (β-radiation)	B1
	left-hand rule (mentioned)	B1
	first finger into page **and** thumb downwards	B1
	current opposite to motion **or** particles negative (and so are beta)	B1
10(a)(iii)	beta (β-radiation) completely absorbed	B1
	some gamma rays absorbed **or** (some) gamma passes through	B1
10(b)	(λ =) $v / f = 3.0 \times 10^{8} / 8.8 \times 10^{19}$	C1
	3.4×10^{-12} (m)	A1

May/June 5054/21

Question 7 · included (a), (b), (c)

(a) Describe the motion of the glider in the first 12 s.

....

....

.... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = ... kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = ... m [2]

Mark scheme — 5054/21 Q7

7(a) wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests B1 7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1 7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1

Question	Answer	Marks
7(a)	wavelength has altered / too large **or** crests in shadow should be bent / quarter circular above or below gap **or** vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap **and** some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz)	A1
7(c)(iii)	1.3 (cm)	B1

May/June 5054/22

Question 5 · included (a), (b)

(a) **(i)** Describe why rotating the tube changes the pressure of the gas in the sealed end.

....

....

.... [1]

(b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is 14 000 kg / m^3.

Atmospheric pressure is 1.0×10^5 Pa.

Calculate the pressure of the gas in the tube.

pressure = ... Pa [3]

Mark scheme — 5054/22 Q5

5(a) seismic S-waves and X-rays B1 5(b)(i) (wavelength =) v / f in any form or (distance =) vt or f = 1 / T or 1 / 0.15 C1 frequency 6.7 (Hz) A1 wavelength 4.0 (cm) A1

Question	Answer	Marks
4(a)(i)	–273 (°C) **and** 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	vibrate / oscillate with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	(E =) mcT in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 **or** 56	C1
	0.48 **or** 0.47 (J / (g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves **and** X-rays	B1
5(b)(i)	(wavelength =) v / f in any form **or** (distance =) vt **or** $f = 1 / T$ or 1 / 0.15	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	straight wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

Question	Answer	Marks
5(b)(iv)	less diffraction **or** less bending	B1

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	B1
	voltmeter V_2 9(.0 V)	B1
6(a)(ii)	(R =) V / I in any form	C1
	36 (Ω)	A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference	B1
	provided the temperature is constant	B1
6(b)	0.35 (A) seen **or** total resistance 36 + 6 / 0.35; 36 + 17(.1); 53(.1) Ω	B1
	voltage (across R) 0.35 × 36 **or** 12.6 (V) seen	B1
	19 V	B1

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1

2022

Oct/Nov 5054/21

Question 4 · included (b)(i)

only frequency definition

(b) Calculate the speed of the car when it reaches B.

speed = ... [3]

Mark scheme — 5054/21 Q4

4(b)(i) number of oscillations / cycles / wavelengths B1 per unit time B1

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Question	Answer	Marks
4(a)	electromagnetic (waves) **or** components of electromagnetic spectrum	**B1**
4(b)(i)	number of oscillations / cycles / wavelengths	**B1**
	per unit time	**B1**
4(b)(ii)	gamma rays	**B1**
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	**B1**
	X-rays pass through flesh but not (to same extent) through bones	**B1**
	image produced by digital / electronic / CCD / CMOS detector	**B1**
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage **or** X-rays show the bone structure rather than foetal development **or** image would not be clear	**B1**

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Question 4 · included (b), (c)

not (a) vacuum exp; not (d) hearing

(b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N / kg.

(c) Describe what is meant by 'a rarefaction'.

....

.... [1]

Mark scheme — 5054/22 Q4

4(b) (they) vibrate M1 (vibration) parallel to / in the direction of propagation / of energy travel A1 4(c) (point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average) B1

Question	Answer	Marks
3(a)	chemical potential energy **and** no other energy	**B1**
3(b)(i)	radiation **or** infrared mentioned	**B1**
	radiation **and** infrared (travel to the man)	**B1**
	energy / infrared / radiation absorbed by the man / pullover **or** travels in straight line **or** travels at the speed of light **or** increases (man's) internal energy	**B1**
3(b)(ii)	(shiny surface) reflects radiation (to man)	**B1**
	less (thermal) energy escapes **or** more (thermal) energy reaches / absorbed by man	**B1**
3(b)(iii)	**EITHER** black (surface) is good absorber / poor reflector of infrared radiation	**B1**
	more (thermal) energy absorbed **or** more thermal energy transferred to man	**B1**
	OR trapped air **and** reduced convection / air poor conductor / air is a good insulator	**(B1)**
	less (thermal) energy lost	**(B1)**

Question	Answer			Marks
4(a)	equipment	e.g.	(evacuable) container **and** (suspended electric) bell / source of sound **and** (air) pump	**B1**
	action	e.g.	ringing / sound (produced) / sound heard **and** evacuate air / vacuum (produced)	**B1**
	observation	e.g.	transmission / hearing of sound ceases / no sound	**B1**

Question	Answer	Marks
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) **or** (point where) pressure / density is lower (than average)	B1
4(d)	(λ =) v / f **or** 1500 / 140 000	C1
	1.1×10^{-2} m **or** 1.1 cm **or** 11 mm	A1

Question	Answer	Marks
5(a)	energy / work done per unit charge **or** $\frac{\text{energy}}{\text{charge}}$ **or** $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply **or** energy used driving charge around a (complete) circuit **or** energy transferred to electrical energy	B1
5(b)	it lasts for a longer time **or** more (electrical) energy available **or** power supply still functions if one cell fails / runs flat **or** can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases **c.a.o.**	B1
	current change **or** voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases **c.a.o.**	B1
5(c)(ii)	(V_1 =) $V_0 \times R_1 / R_0$ **or** 1.5 × 4000 / (4000 + 8000) **or** 1.5 × 4000 / 12 000 **or** $I = V / R$ **or** 1.5 × 8000 / (4000 + 8000) **or** I =1.5 / 12 000 **or** 0.125 mA	C1
	0.50 V	A1

May/June 5054/22

Question 8 · included (a), (b), (c)

not (d) satellite

6 **(a)** Fig. 6.1 shows part of a toy which contains two ring-shaped, permanent magnets. A plastic rod passes through the centre of both magnets.

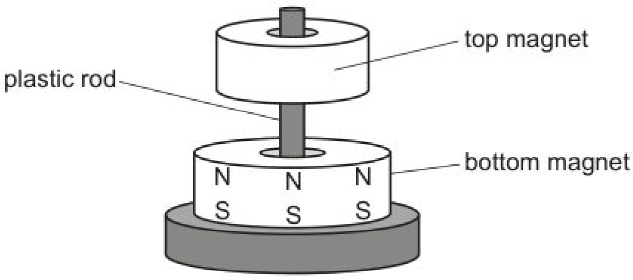

Fig. 6.1

The top magnet can move up and down freely around the plastic rod.

The magnetic poles on the bottom magnet are shown in Fig. 6.1.

(b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0 V, as shown in Fig. 5.2.

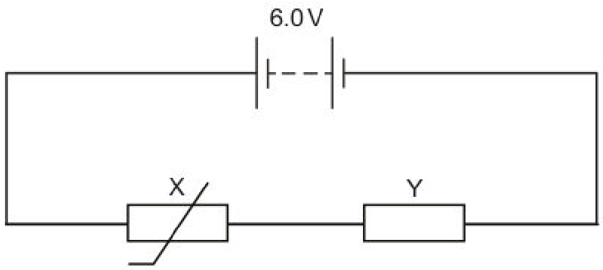

Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is 600 Ω and the current in thermistor X is 0.0020 A.

(b) **(i)** The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

....

.... [1]

(ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed ....

wavelength ....
[2]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

.... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

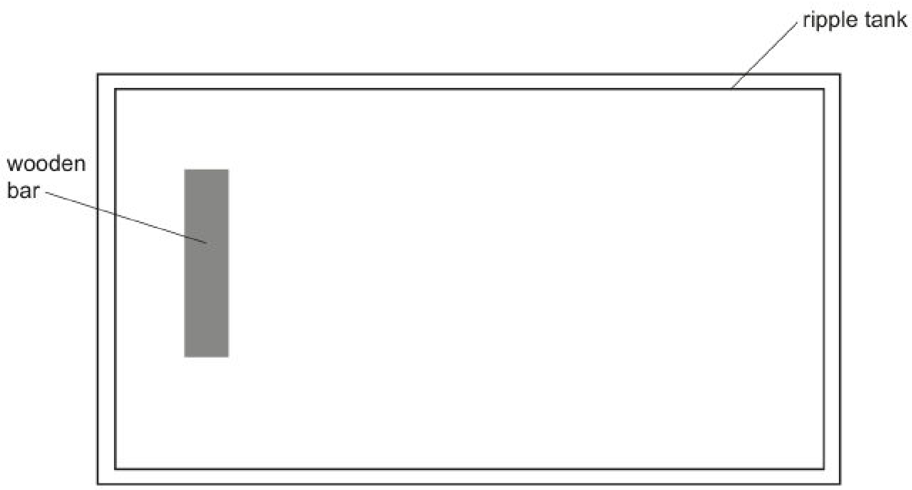

Fig. 8.3

[3]

Mark scheme — 5054/22 Q8

8(a)(i) 3 B1 8(a)(ii) shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change) B1 1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i) B1 8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1 8(c)(i) (piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary B1

Question	Answer	Marks
8(a)(i)	3	**B1**
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	**B1**
	1 wave taking 0.5 s for first wave shown within half a square **or** 1.5 s shown for the number of waves in **(a)(i)**	**B1**
8(b)(i)	move bar up and down more often (per second) / faster **or** make rubber band(s) strong(er) / short(er) / tight(er) / more in number **or** use bar of less mass / weight	**B1**
8(b)(ii)	*speed* remains the same	**B1**
	wavelength decreases	**B1**
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate **or** a shallow(er) / deep(er) area / boundary	**B1**
8(c)(ii)	diagram showing within the ripple tank: • crests parallel to bar hitting boundary with a boundary labelled	**B1**
	• correct refraction of plane crests either towards or away from normal on the same side of the normal **or** no direction change if crests are initially parallel to boundary but with a wavelength change	**B1**
	• different wavelength shown before and after refraction and constant before and after (by eye)	**B1**
8(d)(i)	(t =) distance / speed algebraic or numerical seen in any form	**C1**
	2 × 560(000) seen or used **or** 0.0019 (s)	**C1**
	0.0037 s	**A1**

Question	Answer	Marks
8(d)(ii)	One of (both): • reflect / refract • have same speed (in a vacuum) • travel <u>in a vacuum</u> / don't need a medium • are electromagnetic • are transverse • carry energy / cause heating • have larger wavelength / smaller frequency than visible light	**B1**
8(d)(iii)	less likely to be hacked / intercepted / greater security **or** more channels / better bandwidth possible / more data **or** less likely for transmission to be interrupted **or** cost of <u>satellites</u> saved / <u>satellites</u> expensive	**B1**

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) • two electrons (and ion has 1 electron) • one more electron (than the ion) • no charge (and ion positive) • same number of electrons as protons;	**B1**
9(a)(ii)	alpha particle has (any 1 from) : • no electrons • charge +2 (e) • only protons and neutrons **or** He ion has an electron	**B1**

2021

Oct/Nov 5054/21

Question 8 · included (b)(iii), (b)(v)

only long/trans and v=fλ

(b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

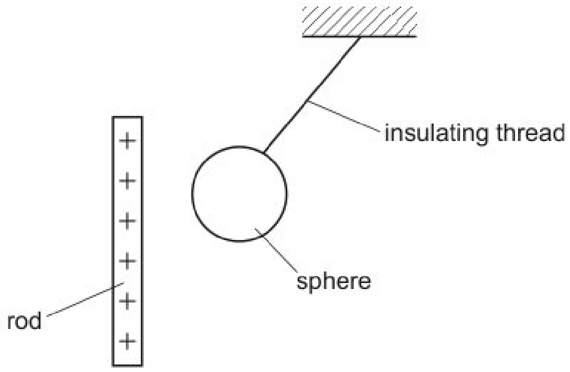

Fig. 5.1

(i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]

(ii) Explain why the sphere is pulled towards the rod.

....

....

....

.... [2]

[Total: 6]

 5054/21/O/N/21

Mark scheme — 5054/21 Q8

8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 © UCLES 2021 Page 9 of 11 Question Answer Marks 8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 © UCLES 2021 Page 9 of 11 Question Answer Marks

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils **or** changing magnetic field in solenoid	**B1**
	an electromotive force / e.m.f. is induced	**B1**
	electromotive force / e.m.f. produces the current	**B1**
7E(b)	negative / opposite / reversed current / deflection / reading	**B1**
	negative e.m.f. / voltage (induced) **or** magnetic field cut in opposite direction **or** magnetic field changes in opposite sense / decreases (owtte)	**B1**
7O(a)	NOR (gate)	**B1**
7O(b)(i)	lower logic gate has an input of logic level 1 (and so its output / Y is 0) **or** the output X has a logic level 1 and so the inputs must both be 0 (Y is an input to the upper NOR gate)	**B1**
7O(b)(ii)	X changes to 0 **and** Y changes to 1.	**B1**
7O(b)(iii)	both X and Y stay at the new values **or** neither X nor Y changes	**B1**
7O(b)(iv)	the output shows which terminal (P or Q) last had the value 1.	**B1**

Question	Answer	Marks
8(a)(i)	magnetic field **or** magnet	**B1**
8(b)(i)	the current (in coil continually) reverses (direction) **or** the force (on the coil continually) reverses	**B1**
	the current (in the coil continually) reverses (direction) **and** the force (on the coil continually) reverses	**B1**

Question	Answer	Marks
8(b)(ii)	cone moves forwards **and** air is compressed / hits (air) molecules	**B1**
	cone moves backwards **and** rarefaction produced / molecules sucked back	**B1**
	(air) molecules vibrate **or** production of compressions and rarefactions continues	**B1**
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	**B1**
	in transverse wave, vibration direction is perpendicular to propagation / energy travel direction	**B1**
8(b)(iv)	(time period =) 0.0050 (s) **or** 1 / 0.0050 **or** 200 (times per second)	**C1**
	400 (times per second)	**A1**
8(b)(v)	(λ =) c / v **or** 340 / 200	**C1**
	1.7 m	**A1**
8(c)(i)	decreases / less loud / quieter	**B1**
	amplitude of sound / vibration decreases **or** (maximum) force on coil decreases	**B1**
8(c)(ii)	(pitch) stays the same / unchanged **and** no change to frequency / time period	**B1**

Question	Answer	Marks
9(a)(i)	molecules with similar / slightly smaller packing density **and** most molecules touching	**B1**
	random arrangement	**B1**
9(a)(ii)	molecules (of gas) further apart	**B1**
	(repulsive) forces between molecules (much) smaller	**B1**

Oct/Nov 5054/22

Question 3 · included (a)

not (b) TIR

(a) State what is meant by a *vector*.

....

.... [1]

Mark scheme — 5054/22 Q3

3(a)(i) it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction B1 change in direction of the ray / refraction shows a change in speed (at the boundary) or sin(i) / sin(r) is ratio of speeds or top of ray slows down first and enters the block first B1 3(a)(ii) (wavelength) decreases B1 (frequency) does not change. B1

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force **or** no resultant force **or** forces balance	B1

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal **or** angle of incidence greater than angle of refraction	B1
	change in direction of the ray / refraction shows a change in speed (at the boundary) **or** sin(*i*) / sin(*r*) is ratio of speeds **or** top of ray slows down first **and** enters the block first	B1
3(a)(ii)	(wavelength) decreases	B1
	(frequency) does not change.	B1
3(b)(i)	(sin(*r*) =) sin(*i*) / *n* **or** sin(45°) / 1.6 **or** 0.71 / 1.6 **or** 0.44 **or** (*r* =) 21(°)	C1
	26(°)	C1
	64°	A1
3(b)(ii)	64° / angle of incidence / *i* is greater than the critical angle / 39° **or** angle of incidence = 64°	B1
	it undergoes total internal reflection	B1

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1

May/June 5054/21

Question 4 · included (a)

not (b) other-topic corrections

(a) The results are shown in **Fig. 2.2**.

Draw a line linking each type of surface with the appropriate meter reading. One line has been drawn for you.

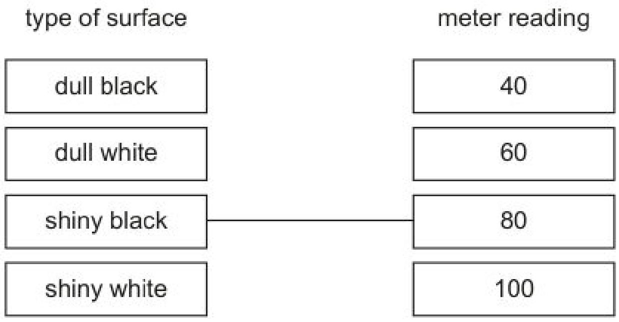

Fig. 2.2

[2]

Mark scheme — 5054/21 Q4

4(a)(i) Fig.4.1 transverse and Fig.4.2 longitudinal B1 4(a)(ii) wave drawn with larger wavelength B1 4(a)(iii) backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement) B1

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PUBLISHED

Question	Answer	Marks
3(a)	(E =) mcT in any form algebraic or numerical	**C1**
	2 × 460 × 450	**C1**
	4.1 ×× 10^5 J	**A1**
3(b)(i)	any one of: • boiling occurs at one temperature / boiling point **or** evaporation occurs at any temperature / below boiling point • boiling occurs within liquid **or** evaporation occurs on the surface • evaporation increased by draughts, more surface area (and boiling does not)	**B1**
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) **or** (molecules) do not have the energy to escape / break bonds	**B1**
	fewer fast / energetic molecules escape **or** (only) fastest / high KE molecules escape	**B1**

Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse **and** Fig.4.2 longitudinal	**B1**
4(a)(ii)	wave drawn with larger wavelength	**B1**
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	**B1**
4(b)	frequency	**B1**
	ultrasound	**B1**
	gamma	**B1**

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May/June 5054/22

Question 10 · included (a)

not (b) spectrum

(a) Describe how the reading on the newton meter is used to find the mass of the brick.

....

.... [1]

Mark scheme — 5054/22 Q10

10(a)(i) 3 (.0) cm B1 10(a)(ii) number of waves per second or 5 / 10 or f = 1 / T C1 2(.0) Hz A1 10(a)(iii) (v =) f λ or s = d / t numerical or algebraic in any form C1 6(.0) cm / s A1 10(a)(iv) frequency stays the same and speed increases B1 5054/22 – Mark Scheme PUBLISHED May/June 2021 © UCLES 2021 Page 15 of 16 Question Answer Marks 10(a)(v) ■ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in- between B1 constant wavelength shown between all refracted wavefronts B1 all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal B1

Question	Answer	Marks
9(c)(i)	(*E* =) *mcT* numerical or algebraic in any form	**C1**
	0.11 × 830 × 5	**C1**
	460 J	**A1**
9(c)(ii)	energy output / energy input	**C1**
	useful energy output / (total) energy input	**A1**
9(c)(iii)	(energy input =) 460 + 5200 **or** 5660 (J) seen **or** 5200 ÷ (energy in **9(c)(i)** + 5200)	**C1**
	92% **or** 0.92	**A1**
9(c)(iv)	(thermal) energy lost / heat lost (from battery to environment)	**B1**
	energy input larger than calculated / 5660 (J) **or** temperature rise should be larger / is smaller than it should be **or** thermal energy / heat produced larger than 460 J / calculated	**B1**

Question	Answer	Marks
10(a)(i)	3 (.0) cm	**B1**
10(a)(ii)	number of waves per second **or** 5 / 10 **or** *f* = 1 / *T*	**C1**
	2(.0) Hz	**A1**
10(a)(iii)	(*v* =) *f λ* **or** s = d / t numerical or algebraic in any form	**C1**
	6(.0) cm / s	**A1**
10(a)(iv)	frequency stays the same **and** speed increases	**B1**

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Question	Answer	Marks
10(a)(v)	$\geqslant$ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case **or** prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out **or** <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected **or** <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1

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2020

Oct/Nov 5054/21

Question 4 · included (a)

not (b) critical angle/TIR

(a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge. The force *P* is 25 N.

Mark scheme — 5054/21 Q4

4(a)(i) it decreases B1 4(a)(ii) it does not change B1

Question	Answer	Marks
3(a)(i)	(Γ =) $Fx_{\perp r}$ **or** 25 × 0.72	**C1**
	18 N m	**A1**
3(a)(ii)	(WD =) $Fx_{\parallel}$ **or** $F\pi r / 2$ or 25 × 4.5 / 4 **or** 25 × 1.13	**B1**
	28 J	**B1**
3(b)	moment due to force *P* is greater	**B1**
	distance of X to hinge is greater than distance from where force *F* acts	**B1**

Question	Answer	Marks
4(a)(i)	it decreases	**B1**
4(a)(ii)	it does not change	**B1**
4(b)(i)	(c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$ **or** $\sin^{-1}(0.625)$	**C1**
	39°	**A1**
4(b)(ii)	reflected ray at Z **and** *i* = *r* **and** no refracted ray	**B1**
	ray strikes vertical face perpendicularly **and** undeviated in air	**B1**

Question	Answer	Marks
5(a)	X, Y and Z to be two (or three) different metals	**B1**
5(b)(i)	(output / thermoelectric) electromotive force / e.m.f. / voltage	**B1**
5(b)(ii)	linearity indicates whether the output (voltage) is directly proportional to something	**B1**
	directly proportional to the temperature difference (between the junctions)	**B1**

2019

Oct/Nov 5054/22

Question 5 · included (a), (b)(i)

not (b)(ii) cleaning

(a) The energy absorbed by solar panels comes from the Sun.

(b) Explain why solar panels are usually black.

....

....

....

.... [2]

[Total: 5]

Mark scheme — 5054/22 Q5

5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 – Mark Scheme PUBLISHED October/November 2019 © UCLES 2019 Page 6 of 10 Question Answer Marks 5(b)(i) (λ =) v / f or 1500 / 42 000 C1 0.036 m or 3.6 cm A1

Question	Answer	Marks
3(a)(i)	(nuclear) fusion (reactions) **or** nuclei fuse	**B1**
	small nuclei / hydrogen nuclei joining together (and release energy) **or** larger nucleus produced	**B1**
3(a)(ii)	infra-red (radiation) **or** ultraviolet (radiation) **or** (visible) light	**B1**
3(b)	black surfaces are (good) absorbers	**M1**
	of (infra-red) radiation / heat / (visible) light / thermal energy	**A1**

Question	Answer	Marks
4(a)	any **two** from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	**B2**
4(b)	molecules escape (from liquid)	**B1**
	fast(est) molecules / molecules with great(est kinetic) energy escape **or** slow(est) molecules / molecules with small(est kinetic) energy remain **or** (average) speed of remaining molecules decreases **or** molecules gain (kinetic) energy and escape	**B1**
4(c)(i)	rate of evaporation increases (with increasing area)	**B1**
4(c)(ii)	rate of evaporation decreases with decreasing temperature **or** takes too much time	**B1**

Question	Answer	Marks
5(a)	vibration / oscillation of particles **or** energy transfer **or** compressions and rarefactions	**B1**
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	**B1**

Question	Answer	Marks
5(b)(i)	(λ =) v / f **or** 1500 / 42 000	**C1**
	0.036 m **or** 3.6 cm	**A1**
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	**B1**
	dirt shaken off	**B1**

Question	Answer	Marks
6(a)(i)	4.0 V **c.a.o.**	**B1**
6(a)(ii)	(V =) 12 – 4.0 **or** 8.0 **or** 4.0 / 700 **or** 5.7×10^{-3}	**C1**
	(R =) V / I **or** 8.0×1 / (4.0 / 700) **or** $700 \times (8.0 / 4.0)$ **or** 12 / (4.0 / 700) **or** 2100 (Ω)	**C1**
	1400 Ω	**A1**
6(b)	resistance of LDR decreases	**B1**
	current increases **or** p.d. across 700 Ω resistor / oscilloscope increases **or** p.d. across LDR decreases	**M1**
	trace moves up the screen	**A1**

Question	Answer	Marks
7(a)(i)	current **and** magnetic field / magnetic flux (experiences a force) **or** the two magnetic fields interact	**B1**
7(a)(ii)	three separate approaches **I** / **II** / **III**	
7(a)(ii)**I**	B to A	**B1**
	left-hand rule mentioned / described	**B1**
	or	

May/June 5054/21

Question 10 · included (a)

not (b) speed of sound exp

(a) (i) Determine the distance between the two towns.

distance = ... [1]

Mark scheme — 5054/21 Q10

10(a)(i) 1.7 (cm) seen or any multiple of wavelength measured on diagram C1 6.4–7.2 cm A1 10(a)(ii) number of waves/cycles/oscillations in one second / unit time C1 number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time or number of oscillations of a source/particle in one second / unit time A1 10(a)(iii) 2.5 Hz B1 10(a)(iv) moved up and down B1 regularly or at constant frequency / 2.5 times a second B1 10(a)(v) speed smaller / decreases and wavelength smaller / decreases B1

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	**C1**
	6.4–7.2 cm	**A1**
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	**C1**
	number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time **or** number of oscillations of a source/particle in one second / unit time	**A1**
10(a)(iii)	2.5 Hz	**B1**
10(a)(iv)	moved up and down	**B1**
	regularly **or** at constant frequency / 2.5 times a second	**B1**
10(a)(v)	*speed* smaller / decreases **and** *wavelength* smaller / decreases	**B1**
10(b)(i)	tape measure **or** trundle wheel **or** microphones	**B1**
	stopwatch **or** timer	**B1**
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	**B1**
	stop timer on hearing sound	**B1**
	measure (large) distance between students	**B1**
10(b)(iii)	both between and including 1000 and 10 000 m/s	**B1**
	solid speed > liquid speed	**B1**

2018

Oct/Nov 5054/21

Question 9 · included (a), (b)

not (c) TIR

(a) On Fig. 3.2, sketch the shape of the graph for loads greater than 7.0 N. [1]

(b) All the loads are removed from the spring. A block of wood is suspended from the end of the spring. The length of the spring increases from 4.9 cm to 13.4 cm.

Determine the weight of the wooden block.

weight = ... [2]

Mark scheme — 5054/21 Q9

9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1 9(b)(i) (f =) v / λ or 0.17 / 0.019 C1 8.9 Hz A1

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

Question	Answer	Marks
9(a)	the vibration direction is parallel to the wave / energy travel direction **or** longitudinal waves have compressions and rarefactions **or** transverse waves have crests and troughs **or** longitudinal waves cannot be polarised	**B1**
9(b)(i)	(f =) v / λ **or** 0.17 / 0.019	**C1**
	8.9 Hz	**A1**
9(b)(ii)	(frequency) stays constant **or** does not change	**B1**
9(b)(iii)	(speed) decreases	**B1**
	(Fig. 9.1 shows that) wavelength decreases **or** refraction is towards normal **or** $i > r$ **or** top of wave (in shaded region) lags behind bottom of wave	**B1**
9(c)(i)	electromagnetic **and** transverse underlined	**B1**
9(c)(ii)	$n = \sin(i) / \sin(r)$ **or** (r =) $\sin^{-1}(\sin(i) / n)$ **or** (r =) $\sin^{-1}(\sin(60°) / 1.6)$	**C1**
	33(°)	**A1**

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ **or** (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$	C1
	39°	A1
	2 total internal reflection	B1
	$\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle	B1
	3 reflected ray at P (correct angle by eye)	B1
	light in air at 30° (by eye) to vertical surface	B1

Question	Answer	Marks
10(a)(i)	same number of protons **or** same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^{4}_{2}\alpha$	B1
	${}^{\dots}_{92}\mathrm{U}$	B1
	${}^{235}_{\dots}\mathrm{U}$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ **or** 5 (half-lives)	C1
	relevant halving **or** $1 / 2^5$ **or** 1 / 32 **or** 6400 / 32	C1
	200 counts / second	A1

Oct/Nov 5054/22

Question 9 · included (b), (c)

not (a) sound vs ultra

(b) When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

(c) The half-life of cobalt-60 is 5.3 years.

(i) State what is meant by *half-life*.

....

....

.... [2]

(ii) When there is a lead sheet between the detector and the sample, the average count rate is obtained from six readings taken at one-minute intervals. The six readings are given in the table in Fig. 6.4.

reading number	1	2	3	4	5	6
count rate / counts/minute	61	46	43	56	49	57

Fig. 6.4

There are reasons for suggesting that the variation in these readings is random and not because the number of cobalt-60 atoms in the sample is decreasing.

State two of these reasons.

1. ....

....

2. ....

....
[2]

Mark scheme — 5054/22 Q9

9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1 9(c)(i) decreases B1

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	B2

Question	Answer	Marks
9(a)	frequency of sound wave small(er) **or** its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium **or** by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction **or** cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled *R*	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. (v =) $f\lambda$ **or** 25 000 × 0.047 / 0.048 / 0.049 **or** 25 000 × 4.7 / 4.8 / 4.9 **or** 25 000 × 47 / 48 / 49	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid **and** no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal **and** slope correct	B1

Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off **or** dislodges / removes dirt	B1

Question	Answer	Marks
10(a)	electrons **c.a.o.**	M1
	towards the ammeter **or** away from the negative terminal **or** towards the positive terminal	A1
10(b)(i)	thermistor **c.a.o.**	B1
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 1.5 + 1 / 6.0$ **or** (R_T =) $R_1R_2 / (R_1 + R_2)$ **or** $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	1.2 (Ω)	C1
	2.5 Ω	A1
10(b)(iii)	(I =) V / R **or** 12 / 2.5	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases **or** total resistance (of circuit) decreases **or** current increases	B1
	voltage (across 1.3 Ω) increases	B1
	trace moves towards top of screen / upwards	B1

May/June 5054/21

Question 11 · included (a), (b)

10 **(a)** **(i)** Describe two differences between boiling and evaporation.

1. ....

....

....

2. ....

....

.... [4]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16 °C
temperature of water after heating = 37 °C
volume of water collected in 1.0 minute = $4.6 \times 10^{-3}\,m^3$
specific heat capacity of water = 4200 J / (kg °C)
density of water = $1000\,kg/m^3$

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass = ...[2]

 5054/21/M/J/18

Mark scheme — 5054/21 Q11

11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1 11(a)(ii) observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object B1 moves up and down B1 11(a)(iii) line joining points M1 that are at the same point in the motion A1 11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	*difference*: Q has a greater / different amplitude or phase	B1
	similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	(f=) 1 / t numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(ii)	2 (λ =) v / f **or** v = dt – numerical or algebraic in any form	C1
	0.060 m	A1

May/June 5054/22

Question 5 · included (b)

only (b)

(b) The object has a constant speed for some of the time.

Mark scheme — 5054/22 Q5

5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	C1
	3.6×10^7 m	A1
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	B1
	3rd column correct microwaves electromagnetic	B1
5(b)(ii)	Layers / molecules / particles / spring coils close together **or** high(er) pressure (than atmospheric)	B1

Question	Answer	Marks
6(a)	**any** two of • ray through middle of lens undeviated • ray parallel to axis passes through focus, 3 cm from lens • ray through focus on left of lens parallel to axis after lens	B2
6(b)	inverted **or** real	B1
6(c)	image size / object size **or** image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) **and** 1.3–1.7	A1
6(d)	projector **or** photographic enlarger	B1

2017

Oct/Nov 5054/22

Question 6 · included (a), (b), (c)

(a) Calculate the pressure on the ground caused by the block.

pressure = ...[2]

(b) State why the total pressure on the ground underneath the block is larger than the value obtained in **(a)**.

....

....[1]

(c) The block is of uniform density.

Mark scheme — 5054/22 Q6

6(a) reflection of sound B1 6(b)(i) decreases B1 6(c) (λ =) c / f or 330 / 3700 C1 0.089 m or 8.9 cm or 89 mm A1 5054/22 – Mark Scheme PUBLISHED October/November 2017 © UCLES 2017 Page 6 of 10 Question Answer Marks

Question	Answer	Marks
5(a)	joining together of (small) nuclei (to make bigger nuclei)	B1
	energy released	B1
	hydrogen (used) **or** helium (produced)	B1
5(b)	electromagnetic radiation / infra-red / light / ultraviolet	B1
	travels through vacuum **or** no medium needed	B1

Question	Answer	Marks
6(a)	reflection of sound	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	(λ =) c / f **or** 330 / 3700	C1
	0.089 m **or** 8.9 cm **or** 89 mm	A1

May/June 5054/21

Question 10 · included (b), (c)(i), (c)(iii)

not (a) mirror; not (c)(ii) hearing

(b) Complete the equation to show the radioactive decay of $^{131}_{53}I$.

$$^{131}_{53}I \rightarrow \,^{......}_{......}Xe + \,^{......}_{-1}\beta + \gamma$$

[2]

(c) When a nucleus of $^{125}_{53}I$ decays, only a gamma ray is emitted.

Suggest why it is better for the doctor to inject the isotope $^{125}_{53}I$ into the patient.

....

....

....

....[2]

Mark scheme — 5054/21 Q10

10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1 10(c)(i) (v =) fλ or 2(000) × 16 C1 32 000 cm / s or 320 m / s A1 10(c)(i) (v =) fλ or 2(000) × 16 C1 32 000 cm / s or 320 m / s A1

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual **or** upright **or** laterally inverted **or** same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase **or** line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	(v =) fλ **or** 2(000) × 16	C1
	32 000 cm / s **or** 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector **or** stated reflector	B1
	detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment	B1

May/June 5054/22

Question 6 · included (a), (b)(i)

not (b)(ii) hearing

(a) Fig. 5.1 shows the arrangement of the molecules in the solid.

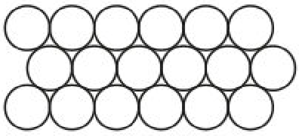

Fig. 5.1

In the space below, draw the arrangement of the molecules in the liquid.

[1]

(b) Complete the table by describing the motion of the molecules in the solid, liquid and gas.

	motion of the molecules
solid	
liquid	
gas	

[3]

Mark scheme — 5054/22 Q6

6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1 6(b)(i) (λ =) v / f or 330 / 3800 B1 0.087 m or 8.7 cm B1

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro **and** in direction of wave **or** has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down **and** at right angles to wave **or** has crests and troughs	B1
6(b)(i)	(λ =) v / f or 330 / 3800	B1
	0.087 m **or** 8.7 cm	B1
6(b)(ii)	not heard **and** because below the range of audible frequencies **or** audible range is 20 – 20 000 Hz **or** too low a pitch / frequency	B1

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) **or** left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical **or** reverses forces (on side AB / CD) **or** keeps forces in same direction for wire on one side	B1
7(b)(i)	(E =)VIt or $2 \times 12 \times 8$ **or** E = Pt **and** P = VI **or** E = VQ **and** Q = It	C1
	190 **or** 192 J	A1
7(b)(ii)	73% **or** 0.73	B1

2016

Oct/Nov 5054/21

Question 5 · included (a), (b)

not (c) production/vacuum

(a) Explain what is meant by the term *critical angle*.

....

....

....

....[2]

(b) The lamp sends light towards the surface of the pool.

Fig. 4.1 shows three rays of light that are at 30°, 60° and 90° to the horizontal.

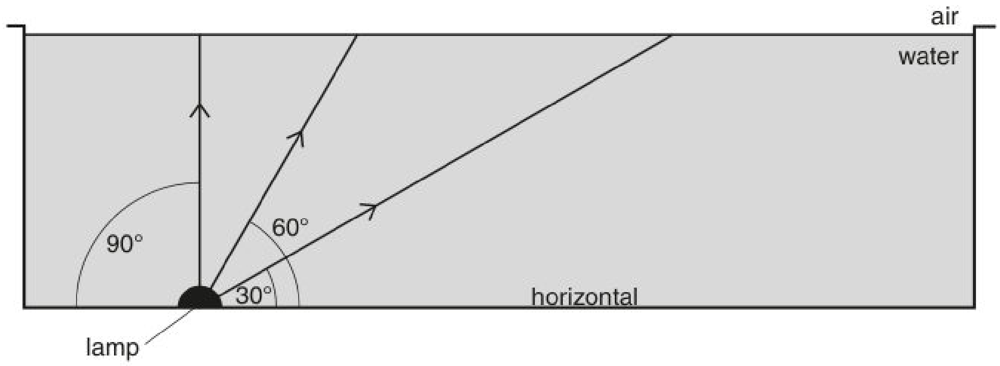

Fig. 4.1

On Fig. 4.1, draw the path taken by each of the three rays after they strike the surface of the water. [3]

Mark scheme — 5054/21 Q5

5	(a)		number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b)	(i)	(λ =)c / f **or** 330 / 2200 0.15 m	C1 A1	
		(ii)	**1** no change **and** **2** increases	B1	
	(c)	(i)	**1** loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) **2** compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1 B1	
		(ii)	fewer / no molecules / particles **and** less / no energy / vibration transferred	B1	[8]
6	(a)	(i)	X N-pole Y S-pole **and** Z N-pole	B1 B1	
		(ii)	they touch / move towards each other **and** opposite poles attract	B1	
	(b)		any sensible use: starting-motor circuit; with a logic gate; nuclear power station corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1 B1	[5]
7	(a)	(i)	supplies the (mains) e.m.f. / voltage	B1	
		(ii)	to complete the circuit / is at 0 V	B1	
	(b)	(i)	the circuit / supply is cut / broken **or** current stops fuse melts / blows / burns	B1 B1	
		(ii)	live wire when it cuts the circuit / melts no part of the appliance is live / no shock	B1 B1	[6]
					[45]

Oct/Nov 5054/22

Question 10 · included (a)

only v=fλ

(a) Calculate the speed of the skier after 5.0 s.

speed = ... [2]

Mark scheme — 5054/22 Q10

		(ii)	1	cylinder is magnetised (by induction) top (of cylinder) is an S-pole unlike poles attract **or** S-pole attracts N-pole	B1 B1 B1	
			2	it does not (remain in contact**) and** iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
						[15]

10	**(a)**	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =)c /f **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$ 7.0×10^{-7} m	C1 A1	[3]
	(b)	**(i)**	decreases	B1	
		(ii)	sin(i) = n × sin(r) **or** 1.5 × sin(30°) **or** 0.75 49°	C1 A1	
		(iii)	41°	B1	[4]
	(c)	**(i)**	dispersion at **both** surfaces **and** refractions in correct direction violet / blue light below the red light shown	B1 B1	
		(ii)	spectrum **or** band of (continuous) colours **or** colours of rainbow red, orange, yellow, green, blue, (indigo, violet)	B1 B1	
		(iii)	**1** X marked above red **2** it is / black surfaces are good absorbers (of IR radiation)	B1 B1	[6]

(d)	intruder / human being emits IR		IR beam broken		IR reflected	B1	
		or		**or**			
	intruder warm **or** IR detected		does not reach detector		change detected	B1	[2]
							[15]

May/June 5054/21

Question 5 · included (a), (b)

(a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

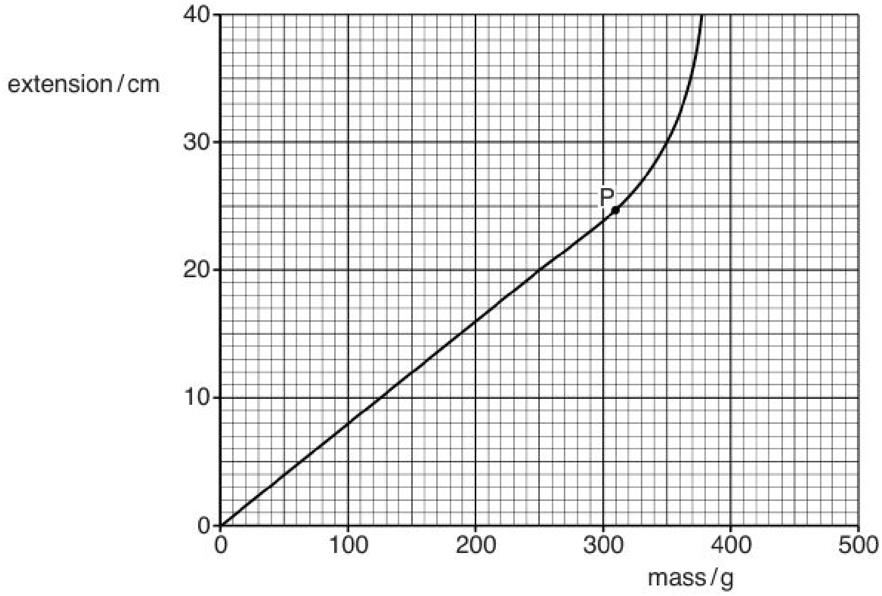

Fig. 2.2

(b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

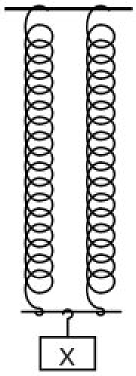

Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

....

....

....

....[2]

Mark scheme — 5054/21 Q5

(b)	**(i)**	**1**	difference between smallest and largest temperature **or** from 0 to 100 °C	B1
	(i)	**2**	small/moderate distance between (thermometer) marks **or** for a given temperature change there is a small expansion of liquid / distance (along scale) / change in thermometric property **or** cannot measure small temperature difference / change	B1
	(ii)		• use liquid that expands more • smaller bore / thinner tube • more mercury (in bulb) **or** use larger bulb	B1

5 **(a)** *sound:* along or parallel (to transfer of energy or wave) **and** longitudinal — B1

water: perpendicular **and** transverse — B1

(b) **(i)** 0.29 – 0.28 m — B1

(ii) time / period for one wave(length) / cycle constant
or each oscillation / cycle takes one second — B1

6 **(a)** angle of incidence — B1

smallest angle for light to be totally internally reflected
or largest angle (of incidence) for ray to be refracted / emerge
or when light emerges along surface
or when angle of refraction is 90° — B1

(b) **(i)** n = 1 / sinC algebraic or numerical — C1
2.5 **or** 2.46 **or** 2.458(59) — A1

(ii) *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface — B1

right hand diagram reflected horizontal ray (by eye) — B1

right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye) — B1

7 **(a)** (current in coil) creates magnetic field
or current is at right angles to magnetic field (of permanent / cylindrical magnets) — C1

(b) into and out of magnet
or left and right
or backwards and forwards — B1

current is one way then reverses (so reverses force) — B1

Question 7 · included (c)

only (c)

(c) The loudspeaker produces sound of frequency 500 Hz. The speed of sound is 320 m/s. Calculate the wavelength of the sound produced.

wavelength = ...[2]

Mark scheme — 5054/21 Q7

	(b)	**(i) 1**	difference between smallest and largest temperature **or** from 0 to 100 °C	B1
		(i) 2	small/moderate distance between (thermometer) marks **or** for a given temperature change there is a small expansion of liquid / distance (along scale) / change in thermometric property **or** cannot measure small temperature <u>difference / change</u>	B1
		(ii)	• use liquid that expands more • smaller bore / thinner tube • more mercury (in bulb) **or** use larger bulb	B1
5	**(a)**		*sound:* along or parallel (to transfer of energy or wave) **and** longitudinal	B1
			water: perpendicular **and** transverse	B1
	(b)	**(i)**	0.29 – 0.28 m	B1
		(ii)	<u>time / period</u> for one wave(length) / cycle constant **or** each oscillation / cycle takes one second	B1
6	**(a)**		angle of incidence	B1
			smallest angle for light to be totally internally reflected **or** largest angle (of incidence) for ray to be refracted / emerge **or** when light emerges along surface **or** when angle of refraction is 90°	B1
	(b)	**(i)**	n = 1 / sinC algebraic or numerical 2.5 **or** 2.46 **or** 2.458(59)	C1 A1
		(ii)	*left hand diagram* ray refracts away from normal and emerges into air at bottom left surface	B1
			right hand diagram reflected horizontal ray (by eye)	B1
			right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye)	B1
7	**(a)**		(current in coil) creates magnetic field **or** current is at right angles to magnetic field (of permanent / cylindrical magnets)	C1
	(b)		into and out of magnet **or** left and right **or** backwards and forwards	B1
			current is one way then reverses (so reverses force)	B1

May/June 5054/22

Question 10 · included (b)

only CRO amplitude/frequency

(b) Calculate the deceleration of the car at $t = 3.0\,s$.

deceleration = ...[2]

Mark scheme — 5054/22 Q10

	(c)		current is (directly) proportional to voltage **or** voltage / current is a constant law holds for constant physical conditions / constant temperature / constant pressure / for metals	B1 B1
	(d)	**(i)**	(directly) proportional **or** (R) ∝ 1	B1
		(ii)	inversely proportional **or** (R) ∝ 1 / A	B1
	(e)		1st band orange 2nd and 3rd bands both black	B1 B1
10	**(a)**	**(i)**	B – anode D – filament **or** heater E and F–Y plates or X plates in either order	B1 B1 B1
		(ii)	**1** attract electrons **or** gives electrons speed / K.E.	B1
			2 heats up cathode **or** gives electrons energy to escape (metal / cathode) **or** causes / allows thermionic emission	B1
		(iii)	kinetic energy to light **or** electrical energy to light	B1
		(iv)	voltage/charge is applied to the X-plates / vertical plates **or** turn on time base (steadily) increasing voltage / charge applied to plate(s) **or** saw tooth voltage applied **or** electrons attracted / repelled by plate(s) or by the electric field between them	B1 B1
	(b)	**(i)**	**1** 1(.0) V	B1
			2 one wave 1.3–1.4 squares **or** 3 waves in 4 squares 2.6–2.8 ms	C1 A1
			3 (f =) 1 / T numerical or algebraic 345–400 Hz	C1 A1
		(ii)	smaller amplitude shown larger period shown	B1 B1

2015

Oct/Nov 5054/21

Question 7 · included (a), (b)

not (c) ultrasound use

(a) Define *boiling point*.

....

.... [1]

(b) **(i)** Describe the molecular structure of a liquid.

....

....

....

....

.... [2]

Mark scheme — 5054/21 Q7

5	**(a)**		**temperature** at which a liquid becomes a gas	B1	
	(b)	**(i)**	molecules close together / touching **or** closer than in gas	B1	
			randomly arranged **or** irregular structure	B1	
		(ii)	to separate / increase the distance between molecules	B1	
			work done against (intermolecular) forces **or** supply p.e. **or** break bonds	B1	[5]
6	**(a)**		**distance** from (optical) centre to focal point (principal focus)	B1	
	(b)	**(i)**	**both** Fs correctly positioned at ±1 mm	B1	
		(ii)	**two** of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (**ign.** arrows)	M2	
			X at crossing point of rays	A1	
		(iii)	3.4–3.8 cm	B1	[6]
7	**(a)**		at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction	B1	
	(b)	**(i)**	$v = f\lambda$ **or** in words	B1	
		(ii)	larger **and** because the frequency is lower	B1	
	(c)		states one use (e.g. prenatal scanning)	B1	
			basic idea (e.g. ultrasound reflects off foetus)	B1	[5]
8	**(a)**	**(i)**	$(I =)V/R$ **or** 12 / (6000 + 2000) **or** 12 / 8000 **or** 12 / 2000 **or** 12 / 6000 **or** in **(ii)** $(V =)IR$ **or** 0.0015 × 6000	C1	
			1.5 mA	A1	
		(ii)	9.0 V	B1	
	(b)		(reading / it) increases	B1	
			resistance of LDR falls	B1	
	(c)		light meter / sensor **or** automatic light switch **or** something sensible	B1	[6]
					[45]

May/June 5054/21

Question 4 · included (a), (b), (c)

(a) State the unit in which pressure is measured.

....[1]

(b) Fig. 3.2 shows the axes for a graph of pressure against volume for the air in the syringe.

One point is plotted on the graph at pressure of P_0 and volume V_0.

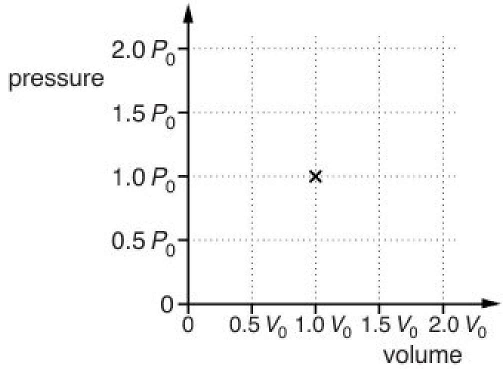

Fig. 3.2

The temperature of the air is kept constant.

On Fig. 3.2,

(c) More metal weights are placed on top of the syringe.

Explain how the molecules of air inside the syringe are able to support more metal weights.

....

....

....

....

....

....

....[3]

Mark scheme — 5054/21 Q4

1	(a)	(i)	60 m	B1
		(ii)	12 s	B1
	(b)	(i)	straight line from origin to 200 m at 40 s any line straight or curved from (40,200) to (60,500)	B1 B1
		(ii)	$s = d / t$ or 500 / 60 8.3 m / s	C1 A1
2	(a)	(i)	force moves through a distance (in same direction)	B1
		(ii)	chemical (potential) energy	B1
	(b)	(i)	480 Nm	B1
		(ii)	attempt to apply moments with two forces and distances 400 N	C1 A1
3	(a)		Pa **or** N / m^2 **or** cm of mercury **or** atmosphere(s)	B1
	(b)		correct points plotted at ($0.5V_0$, $2P_0$) and ($2V_0$, $0.5P_0$) curve through points of decreasing gradient	B1 B1
	(c)		molecules hit sides / piston	B1
			more molecules hit per second / hit more frequently molecular impacts create large(r) **force** (upwards on piston)	B1 B1
4	(a)		oscillate / vibrate stated or described transverse movement described	B1 B1
	(b)		0.40 m	B1
	(c)	(i)	$v = f\lambda$ or (f =) v / λ or 2 / (b) 5.0 Hz	C1 A1
		(ii)	clear attempt to draw wave moved along 0.20 m to right	B1
5	(a)		sin i / sin r or sin 50 / sin 30 1.5(321)	C1 A1

May/June 5054/22

Question 9 · included (a), (b), (c), (d)

(a) Explain how rubbing causes the balloon to become negatively charged.

....

....

.... [2]

(b) Explain why the hair is pulled towards the balloon.

....

....

....

.... [2]

(c) Explain why it is important that the balloon is made from an electrical insulator.

....

.... [1]

(d) State one example where static electricity is useful.

....

.... [1]

Mark scheme — 5054/22 Q9

7	(a)	to provide a complete circuit (with live) **or** to pass current back to mains **or** provide a return path for the current	B1
	(b)	current / charge / electrons flow to earth / earth wire / ground (when live touches case)	B1
		fuse melts / blows **and** disconnects circuit / cuts live / stops current	B1
	(c)	doubly insulated **or case / body** made of plastic / insulator / not made of metal **or** user cannot touch metal	B1
	(d)	(circuit breaker) • turns off / acts fast(er) • can be reset • easy to see it has tripped / switched • can detect small difference between live and neutral currents / small (leakage) current to earth	B1
8	(a)	left column both 1	B1
		right column 0 **and** 1	B1
	(b)	(at least one of the atoms) contain same number of electrons and protons **or have** 1 electron and 1 proton	B1
		charge on electron and proton opposite **or** electron negative **and** proton positive **or** charge on electron neutralises/cancels/balances proton charge	B1
		neutrons have no charge	B1
9	(a)	number of waves (that pass a point) **or** number of oscillations (passing a point)	M1
		in unit time **or** per second **or** in 1 second	A1
	(b) (i)	1.5 cm	B1
	(ii)	$(v =)f\lambda$ **or** 5×1.5 seen **or** $(s=)d/t$ **and** $f = 1/t$	C1
		7.5 cm / s	A1
	(c) (i)	wavelength decreases	B1
		travels a shorter distance in the same time **or** frequency stays the same (and $v = f\lambda$)	B1